

IZT Race Technology

System Architecture Development Specification

Version 1.0 Draft

1. Purpose

Dokumen ini mendefinisikan arsitektur sistem untuk platform IZT Race Technology yang terdiri dari:

- Timing Hardware
- Timing Decoder Software
- Timing API
- Race Management System
- Participant Management System
- Live Leaderboard
- Result Calculation Engine

Tujuan utama sistem adalah menyediakan platform race timing yang scalable, real-time, audit-friendly, dan dapat digunakan untuk berbagai jenis event olahraga.

2. System Design Principles

Principle #1 - Raw Data Must Never Be Lost

Data hasil pembacaan RFID merupakan sumber data utama.

Seluruh pembacaan EPC wajib disimpan sebagai raw timing data.

Raw timing tidak boleh dimodifikasi oleh operator.

Principle #2 - Hardware Does Not Know Participants

Timing hardware maupun decoder software tidak menyimpan data peserta.

Timing system hanya mengetahui:

- EPC
- Timestamp
- RSSI

- Reader ID
- Device ID

Data peserta hanya berada pada platform web.

Principle #3 - Race Results Are Derived Data

Hasil lomba bukan sumber data utama.

Hasil lomba dihitung berdasarkan:

- Raw Timing Data
- Participant Mapping
- Event Configuration
- Race Rules

Result dapat dihitung ulang kapan saja.

Principle #4 - Multi Event Support

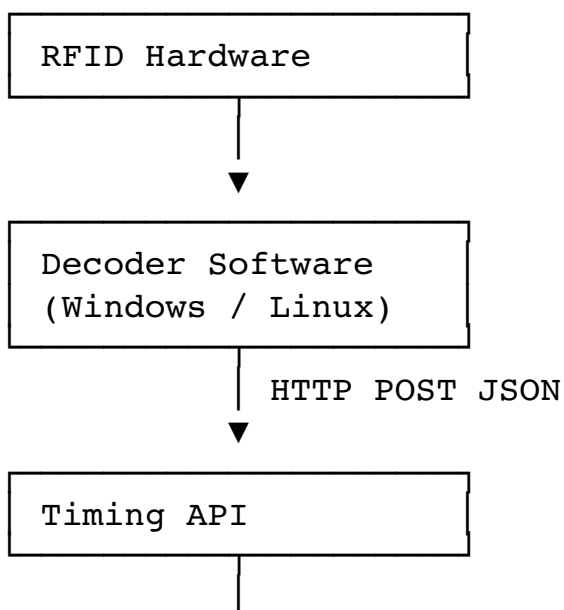
Satu platform harus dapat menjalankan banyak event secara bersamaan.

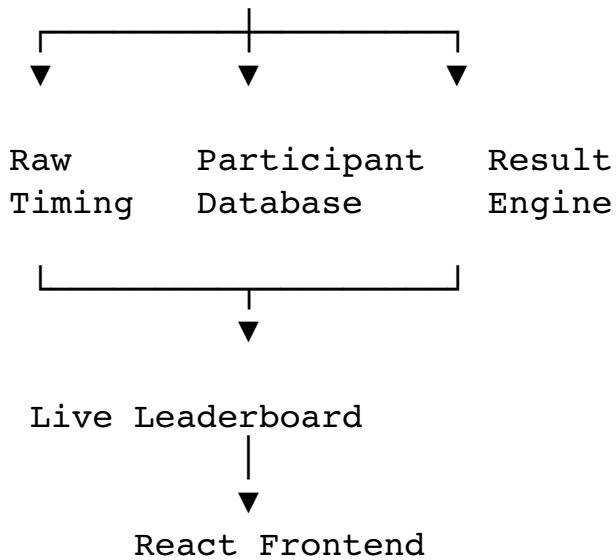
Contoh:

- Kostrad Run 2026
- Bandung Marathon 2026
- BRI Fun Run 2026

tanpa saling mempengaruhi data.

3. High Level Architecture





4. Component Responsibilities

RFID Hardware

Responsibilities:

- Read RFID tags
- Generate EPC
- Generate RSSI
- Send reads to decoder

Must NOT:

- Store participant data
- Calculate ranking
- Calculate race result

Decoder Software

Responsibilities:

- Receive EPC reads
- Buffer data
- Retry failed transmissions
- Send timing data to API

Must NOT:

- Store participant profiles
- Calculate race result
- Calculate leaderboard

Timing API

Responsibilities:

- Receive timing data
- Validate requests
- Store raw timing
- Trigger result processing

Must be stateless.

Race Management System

Responsibilities:

- Event setup
- Participant management
- Category configuration
- Checkpoint configuration
- Result publication

Live Leaderboard

Responsibilities:

- Display ranking
- Display checkpoint passage
- Display participant status
- Receive realtime updates

5. Data Flow

Step 1

RFID Tag Detected

Hardware produces:

```
{  
  "epc": "E20000123456789012345678",  
  "rssi": -52  
}
```

Step 2

Decoder receives read.

Decoder adds:

```
{
  "timestamp": "2026-06-14T08:30:15.123Z",
  "readerId": "CP1",
  "deviceId": "IZT-UHF-001"
}
```

Step 3

Decoder sends payload.

```
{
  "eventId": "kostrad-run-2026",
  "deviceId": "IZT-UHF-001",
  "readerId": "CP1",
  "epc": "E20000123456789012345678",
  "rssi": -52,
  "timestamp": "2026-06-14T08:30:15.123Z"
}
```

Step 4

Timing API stores payload into:

`raw_timings`
table.

Step 5

Result Engine processes timing.

System finds participant mapping.

EPC
↓
Participant
↓
Bib Number
↓
Category

Step 6

Leaderboard updates in realtime.

6. API Specification

Endpoint

POST /api/v1/timing/raw

Headers:

Authorization: Bearer <API_KEY>

Content-Type: application/json

7. Payload Specification

Required Fields

```
{
  "eventId": "kostrad-run-2026",
  "deviceId": "IZT-UHF-001",
  "readerId": "CP1",
  "epc": "E20000123456789012345678",
  "timestamp": "2026-06-14T08:30:15.123Z"
}
```

Optional Fields

```
{
  "rssi": -52,
  "antenna": 1,
  "battery": 87,
  "temperature": 32.4
}
```

8. Database Design

events

field	type
id	uuid
name	varchar
start_date	datetime

status	varchar
--------	---------

participants

field	type
id	uuid
event_id	uuid
bib_number	varchar
full_name	varchar
category	varchar

participant_tags

field	type
id	uuid
participant_id	uuid
epc	varchar

One participant may have multiple tags.

raw_timings

field	type
id	bigint
event_id	uuid
device_id	varchar
reader_id	varchar
epc	varchar

timestamp	datetime
rsi	int

Raw timing records are immutable.

processed_timings

field	type
id	bigint
participant_id	uuid
checkpoint_id	uuid
timing	datetime
lap	int

Generated by Result Engine.

9. Duplicate Detection

Duplicate detection must occur before result calculation.

Default rule:

Same:

- Event
- EPC
- Reader

within:

3 seconds

Store:

First Read Only.

Configuration must be adjustable.

10. Result Calculation Engine

Responsibilities:

- EPC Matching
- Checkpoint Validation
- Lap Detection
- Ranking Calculation
- Category Ranking
- Overall Ranking
- Net Time
- Gun Time

Result Engine must support recalculation.

11. Offline Mode

Decoder must support:

Local Queue

Store unsent timing records.

Retry Logic

Retry every:

5 seconds
until successful.

Guaranteed Delivery

Timing records must not be discarded.

12. Real-Time Communication

Recommended:

SignalR

Alternative:

WebSocket

Events:

TimingReceived

ParticipantPassed

LeaderboardUpdated
RaceFinished

13. Security

Each event has:

Event ID

API Key

Authentication required for all timing submissions.

Rate limiting should be enabled.

All communication must use HTTPS.

14. Scalability Target

Minimum target:

Participants:

10,000+

Concurrent viewers:

5,000+

Timing reads:

500 reads/sec

Leaderboard latency:

< 1 second

API response:

< 200 ms

15. Future Roadmap

Planned support:

- Marathon
- Road Run
- Trail Run
- Cycling
- Duathlon
- Triathlon
- OCR

- Corporate Challenge

Future integrations:

- Strava
- Garmin
- Coros
- Apple Health
- Google Fit

End of Document

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